

PRP2 19-Day Project Plan

Fine-Tuning for Guarani Mbya: Low-Resource Writing Assistant

Compressed 3-week realistic roadmap | 19 calendar days

Current situation: Decision Log 1 is already completed: why this project was chosen. The remaining work is to create a complete, assessable fine-tuning workflow: dataset strategy, baseline model test, LoRA fine-tuning, evaluation, simple demo, ethical reflection, Git evidence, and portfolio documentation.

1. Main Goal and Success Definition

Main goal: Build a small prototype writing assistant for a low-resource language setting and evaluate whether a fine-tuned model gives useful suggestions or mainly memorizes the training data.

Technical result: Cleaned dataset, baseline outputs, LoRA fine-tuning notebook/script, trained adapter or saved outputs, and a simple Streamlit demo.

Research result: Baseline versus fine-tuned output comparison, plus memorization/generalization checks on held-out examples.

Ethical result: A clear explanation that the project is a prototype and that sensitive Indigenous language data should not be used without consent and community control.

Portfolio result: 5-7 meaningful decision logs, Git commits, screenshots, training logs, evaluation tables, and a final report connected to Learning Outcomes.

2. Scope Rules for 19 Days

- Do not try to build a perfect language product. Build a small, complete, responsible workflow.
- Use Guarani Mbya as the ethical and real-world inspiration, but use only safe public data or simulate low-resource conditions if real data is not clearly safe.
- Start with 1,000-3,000 examples. Only expand if the pipeline works early.
- Use a small model first. GPT-2 small or another lightweight open model is enough for a prototype.
- Use LoRA before considering QLoRA. LoRA is easier to implement, explain, and document.
- Keep the Streamlit demo simple. Evaluation is more important than UI polish.

3. Compressed 3-Week Roadmap

Period	Main focus	Output by end of period	Main LOs
Week 1 Days 1-6	Scope, dataset, model feasibility, repository setup	Dataset decision, baseline test, repo structure, DL2/DL3 drafts	LO1, LO2, LO6, LO7
Week 2 Days 7-13	Data cleaning, train/test split, baseline outputs, first LoRA run	Clean dataset, split files, baseline outputs, first training logs, pipeline diagram	LO1, LO3, LO4, LO6
Week 3 Days 14-19	Evaluation, memorization checks, demo, final portfolio evidence	Evaluation table, demo screenshots, final report, decision logs, evidence map, Git cleanup	LO2, LO4, LO5, LO6, LO7

4. Day-by-Day Plan

Week 1 - Decide Before Building (Days 1-6)

Goal: Make the project safe, realistic, and evidence-based before training starts.

Day	Task	What to do	Evidence to save
Day 1	Compress scope and set up repo	Write scope_note.md. Create folders: data/raw, data/processed, notebooks, src, demo, docs, results, portfolio. Add README draft.	scope_note.md, repo commit
Day 2	Search and compare dataset options	Compare 2-3 safe dataset strategies. Include source, license/safety, size, quality, and ethical risk.	dataset_options.md or spreadsheet
Day 3	Choose dataset strategy	Pick the safest and most realistic option. Write source note and limitations. Start DL2.	dataset_source.md, DL2 draft
Day 4	Test base model feasibility	Run a tiny generation test with the chosen base model. Check that the model loads and produces output.	baseline_test.ipynb, screenshots
Day 5	Choose model/fine-tuning approach	Compare small model options and decide LoRA settings at a high level. Start DL3.	model_notes.md, DL3 draft
Day 6	Week 1 checkpoint	Review evidence. Confirm final dataset, base model, evaluation idea, and next week tasks.	weekly_reflection_1.md

Week 2 - Build the Training Workflow (Days 7-13)

Goal: Produce a first working fine-tuning workflow, even if the output quality is still weak.

Day	Task	What to do	Evidence to save
Day 7	Clean dataset part 1	Load data, remove empty rows, broken text, very short examples, and obvious duplicates.	01_cleaning.ipynb
Day 8	Clean dataset part 2	Finalize processed dataset. Save before/after counts and examples of removed rows.	processed dataset, cleaning notes
Day 9	Create splits and baseline prompts	Make train/validation/test split. Keep test examples outside training. Create fixed prompts for baseline and evaluation.	train.csv, validation.csv, test.csv, prompts.csv
Day 10	Generate baseline outputs	Run base model on evaluation prompts before fine-tuning. Save outputs in a table.	baseline_outputs.csv
Day 11	Run first LoRA experiment	Train with small settings first. Save config, loss, runtime notes, and sample generations.	trainingnotebook/script, logs
Day 12	Inspect first results	Compare a few baseline and fine-tuned outputs. Note repetition, nonsense output, or early signs of copying.	early_comparison.md
Day 13	Stabilize pipeline and design diagram	Fix broken parts. Create a simple pipeline diagram: data -> cleaning -> split -> base model -> LoRA -> demo -> evaluation.	architecture.png, DL4 draft

Week 3 - Evaluate, Demo, and Finish Portfolio (Days 14-19)

Goal: Make the project assessable. The assessor should be able to see what was built, how it was tested, what worked, what failed, and why the limits matter.

Day	Task	What to do	Evidence to save
Day 14	Baseline vs fine-tuned comparison	Use the same held-out prompts for both models. Compare usefulness, repetition, and fluency/pattern quality.	evaluation_table.xlsx or csv
Day 15	Memorization check	Compare generated outputs with training examples using similarity checks and manual inspection. Mark possible copied outputs.	03_memorization_check.ipynb
Day 16	Build simple Streamlit demo	Input text -> generated suggestion. Add short disclaimer and simple generation settings.	demo/app.py, screenshot
Day 17	Final evaluation examples	Select good output, weak output, and copied/repeated output examples. Explain what each example shows.	evaluation_examples.md
Day 18	Write final documentation	Finish final report, decision logs, ethical reflection, and evidence map. Link evidence to LOs.	final_report, DL5-DL7, evidence_map
Day 19	Final cleanup and submission proof	Clean README, check Git history, remove junk files, add run instructions, export final screenshots.	GitHub repo, final screenshots, final checklist

5. Decision Log Plan

Important: Do not write a decision log for every tiny action. Write logs only for meaningful methodological choices with alternatives, criteria, evidence, and a decision.

Log	Research question	When	Evidence needed	LO
DL1 - Project choice	Which PRP2 project fits my GenAI Engineer development best?	Already done	Role reflection, project comparison, project description	LO1
DL2 - Dataset strategy	Which dataset strategy is safe, realistic, and useful for low-resource fine-tuning?	Week 1	Dataset comparison, source notes, sample rows, ethics notes	LO1 / LO2
DL3 - Model and LoRA approach	Which base model and fine-tuning method fit my hardware, time, and goal?	Week 1	Model load test, baseline generation, runtime/memory notes	LO1 / LO2 / LO3
DL4 - Preprocessing and split design	How should I clean and split the data to reduce leakage and support evaluation?	Week 2	Cleaning notebook, duplicate check, train/validation/test files	LO3 / LO4
DL5 - Evaluation method	How can I test whether the model generalizes or memorizes?	Week 3	Held-out prompts, similarity checks, good/bad/copied examples	LO2 / LO5
DL6 - Demo scope	What demo scope is enough to prove the prototype without overbuilding UI?	Week 3	Streamlit app, screenshots, user flow, limitations	LO3 / LO4
DL7 - Responsible limits	What would be needed before this could be used responsibly in a real community context?	Week 3	Ethical reflection, no-native-speaker limitation, consent/control discussion	LO6 / LO7

6. Evidence Checklist

Evidence item	Why it matters	Where to store it
README.md	Explains project, setup, run steps, and limitations.	Repository root
scope_note.md	Shows what is inside and outside scope for 19 days.	docs/
dataset_options.md	Shows dataset comparison before the decision.	docs/
dataset_source.md	Explains data source, safety, license/usage, and limitations.	docs/
01_cleaning.ipynb	Shows preprocessing actions instead of only claiming the data was cleaned.	notebooks/
train/validation/test files	Supports fair evaluation and reduces leakage risk.	data/processed/
baseline_outputs.csv	Shows what the model did before fine-tuning.	results/
traininglogs/config	Makes fine-tuning traceable and reproducible.	results/training_logs/
fine_tuned_outputs.csv	Shows the result after training.	results/
memorization_check.ipynb	Supports the central question: learning or copying?	notebooks/
evaluation table	Compares baseline and fine-tuned outputs clearly.	results/
demo screenshots	Proves the prototype works visually.	docs/screenshots/
decision logs	Document why important choices were made.	portfolio/decision_logs/
final report	Makes the full workflow understandable to an assessor.	portfolio/

7. Learning Outcome Mapping

Learning Outcome	How this project shows it	Concrete evidence
LO1 - Analysing	Investigating dataset options, model options, fine-tuning method, and project constraints.	DL2, DL3, dataset comparison, baseline tests
LO2 - Advising	Making justified recommendations about dataset safety, model choice, and responsible limits.	Decision logs, final recommendations, ethics section
LO3 - Designing	Designing the fine-tuning pipeline, data split, evaluation method, and demo flow.	Architecture diagram, preprocessing plan, demo design
LO4 - Realising	Building and testing the working fine-tuning workflow and demo.	Code, notebooks, training logs, Streamlit app, Git commits
LO5 - Managing & Controlling	Monitoring model behavior, checking memorization, identifying weak outputs, and improving settings.	Evaluation table, memorization check, iteration notes
LO6 - Professional Standard	Using a structured method, evidence-based decisions, ethical reflection, and Git documentation.	Project plan, decision logs, evidence map, source notes
LO7 - Personal Leadership	Managing the 19-day plan, asking for feedback, reflecting on limits, and adjusting scope realistically.	Weekly reflections, feedback notes, final reflection

8. Risk and Fallback Plan

Risk	What can go wrong	Fallback action
Dataset is unsafe or unclear	Real language data may be sensitive or not clearly allowed.	Use safe public data or simulate low-resource conditions.

Core priority: dataset -> baseline -> fine-tuning -> evaluation -> decision logs -> simple demo

		Document the decision honestly.
Training is too heavy	Model may fail because of memory or time limits.	Use fewer examples, fewer epochs, smaller model, smaller batch size, or LoRA only.
Output quality is weak	Model may produce repetitive or low-quality suggestions.	Still evaluate honestly. Weak results are acceptable if analysis is strong.
High memorization	Model may copy training examples instead of learning patterns.	Show this as a finding and explain why it limits real-world use.
No native speaker review	Language quality cannot be properly verified.	Avoid strong claims about correctness. Focus on workflow and limitations.
Time pressure	Too much UI or documentation polish may weaken core evidence.	Cut UI polish first. Protect dataset, baseline, training, evaluation, and decision logs.

9. Git and Work Rhythm

- Commit after each meaningful step, not only at the end.
- Use simple prefixes: docs, feat, test, experiment, fix, refactor.
- Link important commits in decision logs as evidence.
- Keep large generated files out of Git if needed, but document where they are stored.
- Every week should end with a short reflection and visible evidence, not only unfinished code.

Example commit	Meaning
docs: add 19-day PRP2 project plan	Shows project planning and scope control.
docs: compare dataset options	Supports DL2 dataset strategy.
test: run baseline generation on held-out prompts	Supports model feasibility and evaluation.
feat: implement LoRA fine-tuning pipeline	Shows realization of training workflow.
test: add memorization similarity check	Supports LO5 monitoring and evaluation.
feat: create Streamlit writing assistant demo	Shows the working prototype.
docs: add final evaluation and responsible limits	Supports portfolio completion.

10. What to Cut If Time Becomes Tight

Protect the core evidence first: dataset -> baseline -> fine-tuning -> evaluation -> decision logs. Cut these items first if needed:

1. Do not polish the demo UI. Keep it simple.
2. Do not test many models. One baseline and one fine-tuned version is enough.
3. Do not chase perfect output quality. Honest evaluation is more valuable.
4. Do not write too many decision logs. Five to seven meaningful logs are enough.
5. Do not expand the dataset if the first smaller dataset already supports the workflow.

11. Definition of Done

- A clear scope note explaining what the project does and does not do.
- A safe dataset strategy with source explanation and limitations.
- A cleaned dataset and train/validation/test split.
- Baseline model outputs before fine-tuning.
- A LoRA fine-tuning notebook or script with saved settings and logs.
- Generated outputs from the fine-tuned model.
- A simple Streamlit demo that gives writing suggestions.
- An evaluation section comparing useful, weak, and memorized outputs.
- A responsible ethical reflection explaining consent, community control, and prototype limits.
- 5-7 meaningful decision logs including the already completed project choice log.
- A Git history that shows development steps and experiments.
- A final report or portfolio page connecting evidence to Learning Outcomes.

12. Simple Daily Working Rule

Every project day should create one small visible artifact: a note, dataset file, notebook result, screenshot, commit, evaluation table, or decision log update. If nothing can be shown, the work is hard to assess.

When asking for guidance, ask about the current artifact, not the whole project. Example: "I found these two datasets; which one is safer and easier to justify?" or "This is my baseline output; is it enough evidence for DL3?"

13. Source Documents Used for This Plan

- Guarani Mbya Writing Assistant - Clear Short Project Description
- Fine-Tuning for Guarani Mbya: AI for Endangered Languages project description
- Portfolio Documentation Guide
- GenAI Project Learning Outcomes Guide
- Decision Log Coaching & Documentation Guide
- How I Will Use Git in This Project
- Decision Log - Project Choice
- Previous 4-week PRP2 Fine-Tuning Project Plan